

Dinesh T V

+91 98420 52589 - [Portfolio](#) - dineshveluswamy@gmail.com - [LinkedIn/dineshveluswamy](#) - [GitHub/thatcatfromspace](#)

EDUCATION

PSG College of Technology

M.Sc Theoretical Computer Science **CGPA: 8.10**

Coimbatore, India

Aug. 2022 - Present

TECHNICAL SKILLS

Programming Languages: C++, Python, Javascript

Libraries and Tools: Git, Docker, Jira, Obsidian, Postman, Valgrind

Frameworks: ReactJS, NextJS, ExpressJS, NestJS

Databases: PostgreSQL, MongoDB, MySQL

WORK EXPERIENCE

Data & AI Consultant

Thorogood Associates

May 2025 - Current

- Collaborated with the infrastructure team to migrate applications from **AWS Elastic Beanstalk** to **EKS** instances, optimizing for better **cost monitoring**, **auto-scaling**, and **instance management using Kubernetes**.
- Implemented **Azure DevOps CI/CD** pipelines to automate deployments in the backend and streamline workflows across staging and production environments.
- Currently revamping a **mission-critical** invoicing system, replacing **legacy Excel macros** with a resilient, web-based backend to modernize deprecated systems.

NON-ACADEMIC PROJECTS

- **Keystore** Minimal Redis clone with TCP socket support and RESP protocol parsing.
 - Implemented a **lightweight, in-memory key-value store** in **C++** using **raw TCP sockets** with custom RESP protocol parsing for command handling.
 - Built the core **store using core C++ STL** to support constant-time operations and multi-threaded the program with **worker threads** to handle expiration.
 - Features includes **multithreaded request handling** with **thread pools**, **TTL-based key expiration**, **LRU caching** and **sharding** for distributed scalability.
- **ECGcomp** Real-time ECG compression system for clinical-grade biosignals.
 - Engineered a **real-time signal compression pipeline** using **delta encoding** and **Zstandard (ZSTD)** to enable low-latency, lossless transmission of high-frequency ECG data over a TCP server.
 - Achieved up to **9.7x compression ratio** across clinical datasets, while preserving full waveform fidelity for downstream diagnostics.
 - Deployed on AWS using a modular pipeline with **CRC validation**, **multichannel signal alignment**, and **parallel encoding** stages.

ACADEMIC PROJECTS

- **Flare** Intelligent last-mile driver assignment and delivery optimization system.
 - Designed a **human-centric scoring engine** that evaluates drivers using **7+ real-world factors**, including sustainability, cost-efficiency, traffic, and delivery acceptance history. Achieved an average savings of **\$0.5 per delivery** and **12.1 tons of CO2** emissions per year.
 - Integrated a **logistic regression-based ML model** to dynamically predict offer acceptance and **optimize incentive payouts**.
 - Normalized driver reliability scores using **real-time weather** and **traffic delays** to create a humane, **redemption-based** optimization engine.
- **ForeSight** A scalable big data weather model for weather forecasting.
 - Uses **Apache Kafka** for **real-time weather data** ingestion and **Apache Cassandra** for distributed storage.
 - Incorporates advanced **lag/rolling** features and employs a **Gradient Boosting Machine (GBM)** for accurate 72-hour weather forecasting.

EXTRACURRICULAR ACTIVITIES

- **Secretary, Computational Sciences Association** *March 2024 - Current* Serving as the secretary and an office bearer for my department's association.